

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q11-0014

Mt Vernon 138 kV  
June 22, 2012

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0014. The Interconnection Customer proposes a 164.6 MW wind facility interconnecting at the Mt Vernon 138 kV substation in the Kentucky area. The request was received on February 26, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 11 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$573,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

For the loss of HUBBARDSTOWN to BAKER 345 kV line, the HUBBARDSTOWN to BAKER 138 kV loads to 101.1% of its 504 MVA rating. Project DFAX: 12.7%.

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

### **4. Required Network Upgrades and Cost Estimates**

| Facility                     | Required Upgrade           | Cost Estimate | Allocation to Project |
|------------------------------|----------------------------|---------------|-----------------------|
| BAKER to HUBBARDSTOWN 138 kV | Reconductor line           | \$573,000     | 100% (\$573,000)      |
|                              | Total allocated to project |               | \$573,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.